

CS & IT ENGINEERING



Computer Network

MAC Sub-Layer

Lecture No. - 05



By - Abhishek Sir



Recap of Previous Lecture



Topic

CSMA



Topics to be Covered



Topic

CSMA/CD $\Rightarrow$ V.V. Imp.

Topic

Ethernet

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW





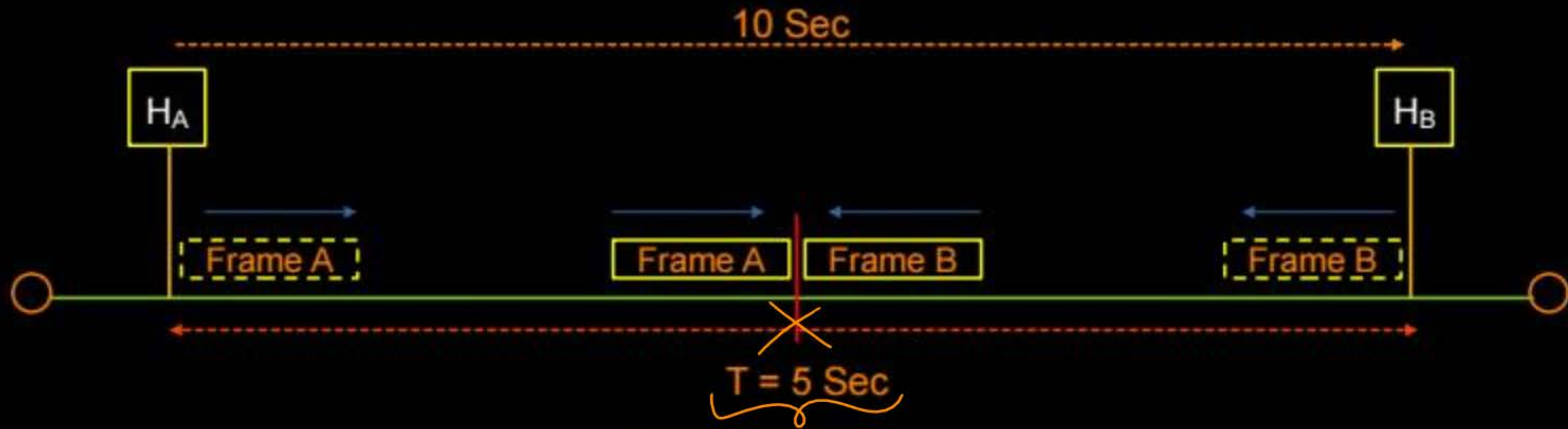
Topic : CSMA



- Carrier Sense Multiple Access
- Sense before transmit
[Sense the channel, before transmission]
- if channel sensed idle : “transmit entire frame”
- if channel sensed busy : “defer transmission”

CASE I :-

Frame Transmission Time = 1 Sec



Host A

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0 \text{ Sec}$

Complete transmission at time $T = 1 \text{ Sec}$

Receive collision signals at time $T = 10 \text{ Sec}$

Host B

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0 \text{ Sec}$

Complete transmission at time $T = 1 \text{ Sec}$

Receive collision signals at time $T = 10 \text{ Sec}$

CASE II :-

Frame Transmission Time = 1 Sec



Host A

Sense channel for transmission at time T = 0

Start transmission at time T = 0 Sec

Complete transmission at time T = 1 Sec

Receive collision signals at time T = 16 Sec

Host B

Sense channel for transmission at time T = 6

Start transmission at time T = 6 Sec

Complete transmission at time T = 7 Sec

Receive collision signals at time T = 10 Sec

CASE III :- Frame Transmission Time = 1 Sec



Host A

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0$ Sec

Complete transmission at time $T = 1$ Sec

Not Receive any collision signals

Host B

Sense channel for transmission at time $T = 10.5$

Transmission defer as it find channel is busy



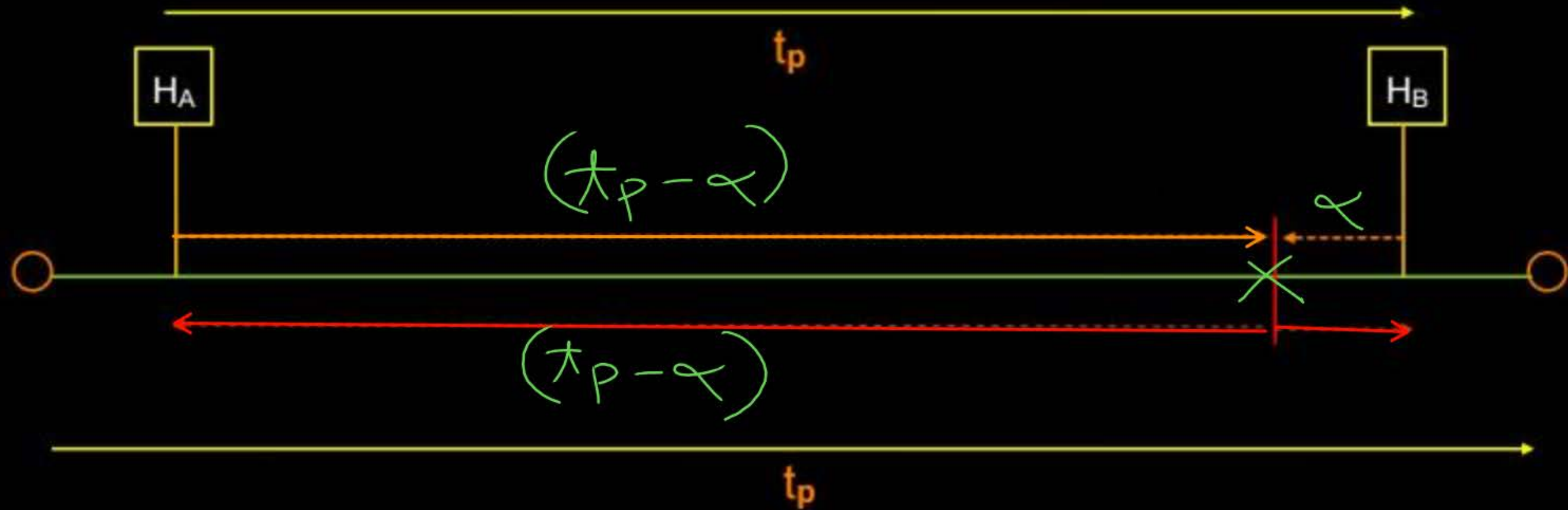
Topic : CSMA/CD



- CSMA with Collision Detection
- Applicable only for wired LAN (Bus topology)
- Sense before transmit
- Sense while (during) transmission ✓
- No any feedback (acknowledgment) from receiver
(Ack does not exist)



Topic : CSMA/CD



$$t_x \geq 2(t_p - \alpha)$$

$$t_x \geq 2t_p$$



Topic : CSMA/CD



- To detect collision,
minimum frame transmission delay should be
greater than equal to (maximum) round trip propagation delay.

frame transmission delay $\geq$ round trip propagation delay

$$t_x \geq 2 * t_p$$

IV
CASE VI:-

$$\text{Frame Transmission Time} = \boxed{20 \text{ Sec}} = [2 t_p]$$

$$t_p = [10 \text{ Sec}]$$



Host A

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0 \text{ Sec}$

It will complete transmission at $T = 20 \text{ Sec}$

Receive collision signals at time $T = 16 \text{ Sec}$

Host B

Sense channel for transmission at time $T = 6$

Start transmission at time $T = 6 \text{ Sec}$

It will [complete transmission] at $T = 26 \text{ Sec}$

Receive collision signals at time $T = 10 \text{ Sec}$



Topic : CSMA/CD



For minimum frame size

$$t_x = 2 * t_p$$

$$\text{minimum frame size} = (2 * t_p) * \text{Bandwidth}$$



#Q. A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network?

- (A) 50 bytes
- (B) 100 bytes
- (C) 200 bytes
- ✓ (D) None of the above

CSMA/CD

$$t_x \geq 2t_p$$

[GATE 2003]

(IIT-M)

Ans: D

Solution :

$$\text{Minimum frame size} = 2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth}$$

$$= 2 * (2 \text{ km} / 2 * 10^8 \text{ m/sec}) * 10^7 \text{ bits/sec}$$

$$= 2 * (2 * 10^3 \text{ m} / 2 * 10^8 \text{ m/sec}) * 10^7 \text{ bits/sec}$$

$$= 200 \text{ bits}$$

$$= 25 \text{ bytes}$$

#Q. A network with CSMA/CD protocol in the MAC layer is running at 1 Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The minimum frame size for this network should be

- (A) 10000 bits
- (B) 10000 bytes
- (C) 5000 bits
- (D) 5000 bytes

[GATE 2004]

11T-D
H.W

#Q. The minimum frame size required for a CSMA/CD based computer network running at 1 Gbps on a 200 m cable with a link speed of 2×10^8 m/s is :

[GATE 2008]

- (A) 125 bytes
- (B) 250 bytes
- (C) 500 bytes
- (D) None of these

ISC

H.W.

#Q. A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.

[GATE 2016]

11SC

H.W.

#Q. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometre) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

[GATE 2015]

IIT-K
H.W.

- (A) 8000
- (B) 10000
- (C) 16000
- (D) 20000

In CSMA/CD :-

Min^m transmission time = $2t_p$
(by any host)

- ① $t_x = 2t_p$
② $t_x > 2t_p$





Topic : Jam Signal

$\Rightarrow$ continuous one's



- Jamming signal 32-bits or 48-bits (min^m length)
- Transmitter transmit jam signal when collision detected
- To inform the other transmitting stations about the collision

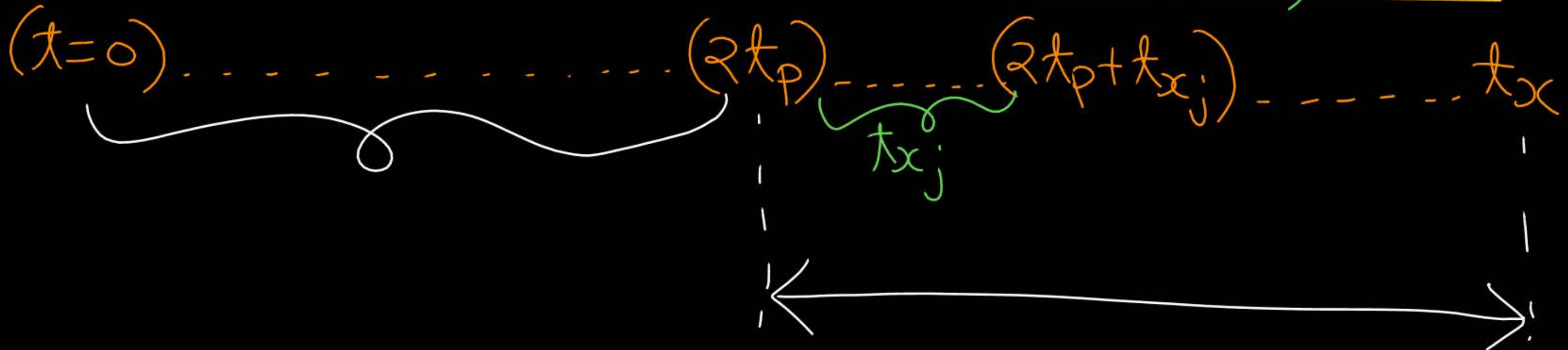
$$t_x \geq [2 * t_p + \underline{t_{xj}}]$$

t_{xj} : Transmission time for Jamming Signal (min^m size)
Not given, consider negligible

$t=0$ $2t_p$ t_x

$\Downarrow$
 Collision
 Detected
 it transmit
 min^m size
 jam signal

Min^m transmission time = $(2t_p + t_{xj})$



#Q. Suppose the round trip propagation delay for a 10 Mbps Ethernet having 48-bit jamming signal is $46.4 \mu s$. The minimum frame size is:

[GATE 2005]

IIT-B

$$2t_p = 46.4 \mu s$$

$$\text{Size of jam signal} = 48 \text{ bits}$$

Ethernet (CSMA/CD)

① Negligible jam signal size
 $[t_x \geq 2t_p]$

② Include jam signal size
 $[t_x \geq (2t_p + t_{x_j})]$

Ans: D

(A) 94 bits

(B) 416 bits

(C) 464 bits

(D) 512 bits

Solution :

$$t_x = [2 * t_p + t_{xj}]$$

$$\text{Minimum frame size} = [2 * t_p + t_{xj}] * \text{Bandwidth}$$

$$= [46.4 \mu\text{s} + (\text{Jam Signal Size} / \text{Bandwidth})] * \text{Bandwidth}$$

$$= [46.4 \mu\text{s} * \text{Bandwidth} + \text{Jam Signal Size}]$$

$$= [46.4 \mu\text{s} * 10\text{Mbps} + 48 \text{ bits}]$$

$$= [46.4 \mu\text{s} * 10^7 \text{ bits/sec} + 48 \text{ bits}]$$

$$= [464 \text{ bits} + 48 \text{ bits}]$$

$$= 512 \text{ bits}$$

$$= 64 \text{ bytes}$$



Topic : Ethernet

$\Rightarrow$ LAN standard



$\rightarrow$ IEEE 802.3

$\rightarrow$ Based on 1 - persistent CSMA/CD



$$t_x \geq 2t_p$$

$$t_x \geq [2t_p + t_{x_j}]$$

#Q. Consider an Ethernet segment with a transmission speed of 10^8 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (*in bits*) required for collision detection is _____.

[GATE-2024, Set-2, 2-Mark]

ISC, H.W.

#Q. Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

- (A) 1
- (B) 2
- (C) 2.5
- (D) 5

[GATE 2013]

IIT-B
(H.W.)



Topic : Ethernet Standard



→ Thick Ethernet (Thicknet) : 10 BASE5 → 500 m

→ Thin Ethernet (Thinnet) : 10 BASE2 → 185 m

→ 10 Mbps Bandwidth (Baseband) ⇒ Digital signal

→ 500 meter / 200 meter (Segment Length) [without Repeater]
185



Topic : Ethernet

⇒ Based on CSMA/CD



→ Minimum frame size : 64 Bytes

→ Maximum frame size : 1518 Bytes

→ Inter-frame gap between frames

→ Ethernet uses Jamming Signal (mostly 48 bit)



2 mins Summary



Topic

CSMA/CD ✓

Topic

Ethernet



THANK - YOU

